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System Of Linear Equations

1.
$$\begin{aligned} x - 3y &= 3 \\ x + y &= 7 \end{aligned}$$

①

The solution to the system of equations above is (x, y) . What is the value of x ?

- A) 2
- B) 4
- C) 5
- D) 6

2. The solution to the given system of equations is (x, y) . What is the value of $x - y$?

①

$$\begin{aligned} 6x &= 30 \\ 3x + 2y &= 8 \end{aligned}$$

- A) -3.5
- B) 1.5
- C) 5
- D) 8.5

3.
$$\begin{aligned} x - y &= 3 \\ y &= 3 - 2x \end{aligned}$$

①

For the system of equations above, what is the value of x ?

- A) 6
- B) 2
- C) 0
- D) -3

4.
$$\begin{aligned} 3m + 5n &= 26 \\ m + n &= 34 \end{aligned}$$

①

In the system of equations above, what is the value of m ?

- A) 18
- B) 36
- C) 72
- D) 144

5.
$$\begin{aligned} 2x - 3z &= -6 \\ 3x + 2z &= 4 \end{aligned}$$

①

For the system of equations above, what is the value of $x + z$?

- A) -2
- B) 0
- C) 1
- D) 2

6.
$$\begin{aligned} 0.25a - 2b &= -4 \\ a + b &= 2 \end{aligned}$$

①

The solution to the system of equations above is the point (a, b) . What is the value of b ?

- A) 2
- B) 3
- C) 4
- D) 5

- 1 7. If (x, y) satisfies the system of equations below, what is the value of x ?

$$\begin{aligned} 3x - 2y &= -1 \\ 10x + 4y &= 50 \end{aligned}$$

A) 3 B) 4 C) 5 D) 6

- 2 8. Max has \$150 to spend on a picnic. He is going to buy cookies and brownies for the picnic. The table below gives the number of brownies and cookies in each box and the price per box.

Food Item	Number in each box	Price per box
Cookies	18	6
Brownies	12	8

Max will order a total of 20 boxes of cookies and brownies. Which system of equations gives a true relationship between the boxes of brownies, B , and boxes of cookies, C , that Max will order?

- A) $C + B = 20$
 $6C + 8B = 150$
B) $C + B = 20$
 $18C + 12B = 150$
C) $C + B = 300$
 $6C + 8B = 150$
D) $6C + 8B = 20$
 $18C + 12B = 150$

- 1 9. If (x, y) is the solution to the system of equations below, what is the value of $y - x$?

$$\begin{aligned} 2x - 3y &= -9 \\ 3x + 2y &= 19 \end{aligned}$$

- A) -2
B) 2
C) 3
D) 8

- 1 10.
$$\begin{aligned} 6x &= y + 8 \\ y &= 96 - 7x \end{aligned}$$

In the system of equations above, what is the value of $x - y$?

- A) -32
B) 32
C) 104
D) 736

- 1 11. The solution to the given system of equations is (p, q) . What is the value of q ?

$$\begin{aligned} 8x + y &= 48 \\ 3x + y &= 12 \end{aligned}$$

A) -9.6 B) -10.8 C) -11.2 D) -11.4

- 2 12. Monique, a manager at a local bakery, is taking all of the \$10 and \$20 bills in the cash register to the bank at the end of her shift. On her way back to work, she lost the deposit slip, but she remembers that there were 19 bills totaling \$260. How many \$10 bills were in the cash register at the end of her shift?

- A) 6
B) 9
C) 11
D) 12

- 1 13. The system of equations below has a solution (x, y) . What is the value of y ?

$$\begin{aligned} \frac{1}{4}x &= 4y - 4 \\ -\frac{3}{4}x + 5y &= -\frac{55}{10} \end{aligned}$$

A) 2.5 B) 5 C) 3.5 D) 4.5

- 2 14. The drama club is selling two types of tickets for the play: student tickets for \$7 each and regular tickets for \$12 each. If the drama club sold a total of 200 tickets and the ticket sales were \$2,025, how many student tickets were sold?

A) 75 B) 50 C) 100 D) 90

- 2 15.
$$\begin{aligned} 4a + 3b &= 22 \\ 3a + 2b &= 16 \end{aligned}$$

In the system of equations above, what is the value of $a + b$?

A) 6 B) 7 C) 9 D) 3

16. At Franky's Surf Shop, the price of a t-shirt is \$20, and the price of a hat is \$17. Javaun spent \$282 on 15 items and only bought t-shirts and hats. How many hats did he buy? (2)
A) 6 B) 4 C) 8 D) 10
17. Kiki's teacher wrote the following riddle on the board: "There are two integers I am thinking of. The difference between the two integers is 21, and the sum of the larger integer and twice the smaller integer is -15 . What is the value of the larger integer?" The answer is: (2)
A) -12
B) -9
C) 9
D) 12
18.
$$\begin{aligned} -4x - 15y &= 78 \\ 3y + x &= -30 \end{aligned}$$
 (2)
The system of equations above is graphed in the xy -plane. At which point do the graphs intersect?
A) $(14, -14)$
B) $(-72, 14)$
C) $(7, -52)$
D) $(228, 66)$
19.
$$\begin{aligned} y &= x^2 - 11x + 14 \\ 2y &= 4 - 6x \end{aligned}$$
 (2)
The system of equations above is graphed in the xy -plane. Which of the following is the x -coordinate of an intersection point of the two equations?
A) 1
B) 4
C) 6
D) 10
20. At the Pacific Beach boardwalk, there is a taco stand that sells carnitas tacos and fish tacos. Carnitas tacos have 50 more calories than fish tacos. If 3 carnitas tacos and 2 fish tacos have a total of 650 calories, how many calories does each fish taco have? (2)
A) 100 B) 75 C) 125 D) 150
21.
$$\begin{aligned} x &= y^2 + 8y \\ x + 2y &= 24 \end{aligned}$$
 (2)
The system of equations above is graphed in the xy -plane. Which of the following is the y -coordinate of an intersection point of the graphs of the two equations?
A) 8
B) 4
C) -2
D) -12
22. A group of 150 people went whitewater rafting. The group took 20 total rafts. The rafts could carry either 6 or 8 people. How many 8-person rafts did the group take? (2)
A) 4
B) 6
C) 10
D) 15
23. The graphs of the equations in the given system of equations intersect at the point (x, y) in the xy -plane. What is a possible value of x ? (2)
$$\begin{aligned} 14x + \frac{1}{2}y &= -28 \\ 2x^2 &= y - 40 \end{aligned}$$

A) -6 B) -8 C) -10 D) -12
24. Jimmy's Deli sells Reuben sandwiches for \$7.35 and brisket sandwiches for \$8.10. Yesterday, the deli made \$759.90 from selling a total of 99 Reuben and brisket sandwiches. How many Reuben sandwiches were sold yesterday? (2)
A) 56 B) 49 C) 63 D) 70

25. The solution to the given system of equations is (a, b) . What is the value of a ?

$$\begin{aligned}a + b &= 17 \\ 3(a - b) &= 33\end{aligned}$$

A) 14 B) 16 C) 18 D) 21

26. The product of 2 integers is 96. The greater integer is 2 less than triple the lesser integer. What is the greater integer?

A) 16 B) 18 C) 20 D) 24

27. $(x - 1)^2 + (y - 2)^2 = 35$
 $x = 2y + 2$

Which of the following could be the x -coordinate of the solution to the system of equations above?

A) $\sqrt{6}$
B) $\frac{\sqrt{30}}{2}$
C) $2\sqrt{6} + 2$
D) $\sqrt{30} + 2$

28. The solution to the given system of equations is $(2, 3)$. What is the value of $a + b$?

$$\begin{aligned}4ax &= by + 12 \\ 2ax - 5by &= 15\end{aligned}$$

A) $7/12$ B) $5/12$ C) $1/2$ D) $3/4$

29. The equations $c = x^2 - y$ and $(x - 3)^2 + (y + 1)^2 = 49$ intersect at the point $(3, y)$. What is the value of c ?

A) -1
B) 1
C) 3
D) 4

30. $4x - 5qy = 36$
 $px + 10qy = 54$

In the given system of equations, p and q are constants. The lines of the two equations intersect at $(4, y)$. What is the value of p ?

A) 23.5 B) 24.5 C) 24 D) 25

31. The linear function f is defined by $f(x) = ax + b$, where a and b are constants. If $f(15) = 850$ and $f(28) = 1,305$, what is the value of $a + b$?

A) 360 B) 240 C) 300 D) 450

32. $8kx + 8my = -8$
 $3xk + 6my = -24$

In the system of equations above, k and m are constants. The system has a solution of $(6, y)$. What is the value of k ?

A) 1 B) 2 C) 3 D) 4

33. The system of equations below intersects at the point $(-3, y)$. What is the value of h ?

$$\begin{aligned}20 + 2y &= 8x \\ y &= x^2 - 16hx - 43\end{aligned}$$

A) $1/4$ B) $1/2$ C) $1/3$ D) 1

34. In the system of linear equations below, c is a nonzero constant. The graphs of the equations are two lines in the xy -plane that intersect at $(-7, a)$. What is the value of a ?

$$\begin{aligned}5y + 7x &= 3c \\ 4y + 9x &= c\end{aligned}$$

A) 20 B) 10 C) 30 D) 40

35. $(a - b) + 12(c + d) = 589$
 $(a - b) + 18(c + d) = 1251$

For the system of equations above, what is the value of $2a - 2b$?

A) -1470 B) -1320 C) -1400 D) -1520

36. James took a trip from Maryland to Boston by car and train. When travelling by car, he averaged 40 miles per hour. When traveling by train, he averaged 60 miles per hour. The trip took him 7 hours and was 400 miles. How far did he travel by train?

A) 120
B) 200
C) 240
D) 360

37.
$$y = x - 18$$
$$y^2 + (x - 14)^2 - 12 = 0$$
For the system of equations above, what is a possible value of y ?

A) $-2 + \sqrt{2}$
B) $\sqrt{2}$
C) $2 + \sqrt{3}$
D) $-2 - \sqrt{6}$

38. The quadratic function f is defined by $f(x) = ax^2 + bx + c$, where a , b , and c are constants. If $f(9) = 251$, $f(3) = 71$, and $c = 35$, what is the value of $f(7)$?

A) 175 B) 150 C) 200 D) 225

39. In the system of equations below, a is a constant. What is the y -value of the solution to the system in terms of a ?

$$\begin{aligned} 3x + 4y &= 6 \\ -5x + y &= a \end{aligned}$$

A) $\frac{30+3a}{23}$
B) $-\frac{30+3a}{23}$
C) $\frac{6+4a}{23}$
D) $-\frac{6+4a}{23}$

40. In a chemistry classroom, when experiments are performed, students are seated at lab tables. If the teacher assigns 4 students to each lab table, 2 additional lab tables will be needed to seat all of the students. If the teacher assigns 8 students to each lab table, there will be 2 extra tables. How many students are in the science class?

A) 32 B) 36 C) 40 D) 42

41. The difference of 2 positive integers is 5. The square of the greater number is 95 more than the square of the lesser number. What is the value of the sum of the two integers?

A) 19 B) 22 C) 25 D) 27

42. In a group of 750 college students, some are premed students, and the rest are business students. There are 150 more women than men and there are 300 fewer business students than premed students. If there are 90 female business students and a man is to be selected at random, what is the probability that the man is a premed student?

A) 0.55 B) 0.45 C) 0.35 D) 0.65

43. For the system of equations below, the lines intersect at the point $(2, 4)$. If $b > 0$, what is the value of b ?

$$\begin{aligned} \frac{1}{2}ay - \frac{1}{2}bx &= 7x \\ \frac{1}{2}ay &= \frac{1}{2}b^2y - 3bx + 5x \end{aligned}$$

A) 4 B) 2 C) 1 D) 8

44. The functions f and g are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $f(x) = 103\left(\frac{1}{1.1}\right)(1.1)^{x+1}$
II. $g(x) = 103(1.44)(1.2)^{x-2}$

A) I only
B) II only
C) I and II
D) Neither I nor II